

ODI Cricket Match Analysis (1971–2024)

IBM SkillsBuild Data Analysis Project

This project analyzes 4,745 historical ODI match records from 1971-01-05 to 2024-03-18, using match, batting, bowling and player information.

1. Introduction

One Day International cricket has evolved considerably since the first ODI era. This project uses historical ODI data to identify performance patterns and changes over time. The analysis focuses on match outcomes, team performance, toss outcomes, batting-first results, scoring trends, and leading individual performers.

2. Objectives

The main objectives are to (1) examine the growth of ODI matches across decades, (2) compare teams by wins and win rates, (3) investigate whether winning the toss is associated with winning the match, (4) analyze batting-first outcomes, (5) identify leading run scorers and wicket takers, and (6) study scoring trends.

3. Dataset Description

The Kaggle dataset contains six CSV files: match-level data, batting cards, bowling cards, fall-of-wickets, partnership cards, and player information. The main match file contains 4,745 records. The supplied snapshot spans 1971-01-05 to 2024-03-18.

File	Purpose
odi_Matches_Data.csv	Match dates, teams, scores, toss, winner and venue
odi_Batting_Card.csv	Runs, balls, fours, sixes and dismissals
odi_Bowling_Card.csv	Overs, runs conceded, wickets and economy
odi_Fow_Card.csv	Fall-of-wicket details
odi_Partnership_Card.csv	Batting partnership details
players_info.csv	Player names and playing information

4. Data Preparation

Dates were converted to datetime values and score fields were converted to numeric values. The dataset has no duplicate Match ID values. 210 matches do not have a recorded Match Winner, so winner-based calculations use 4,535 decisive records. There are 4 missing toss-choice values and 101 missing Team 2 score values. Missing values were not replaced with invented values; they were excluded only from calculations that required them.

5. Exploratory Analysis

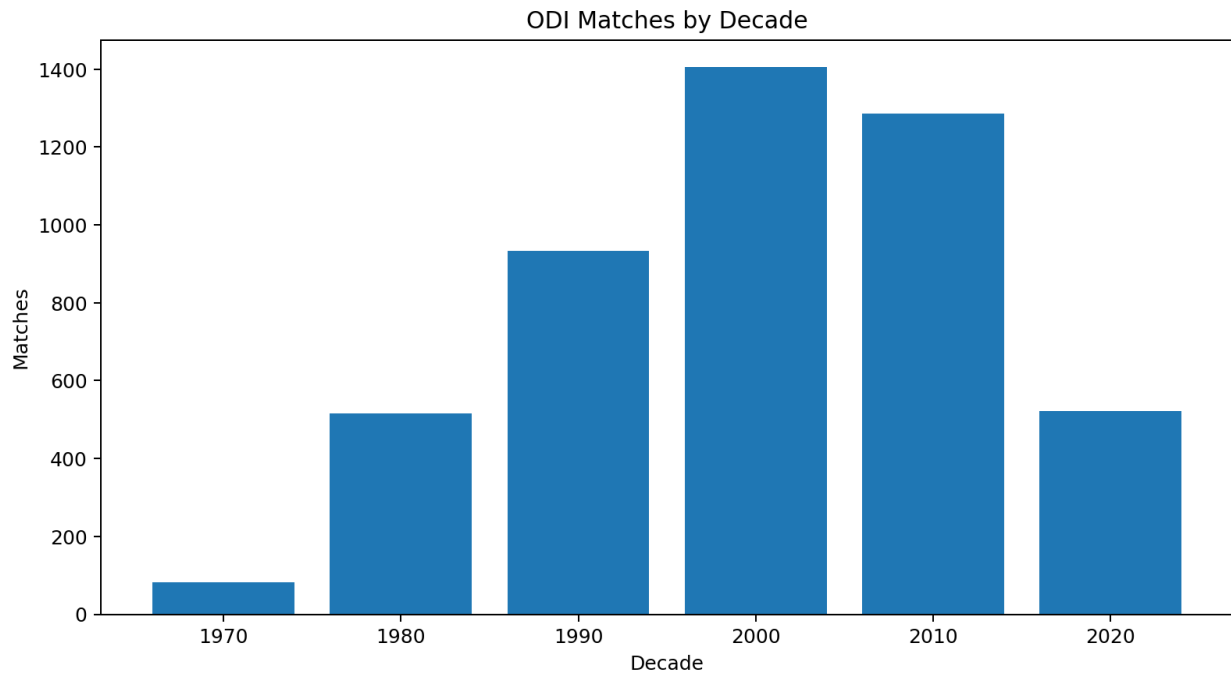


Figure 1. Number of ODI matches represented in each decade.

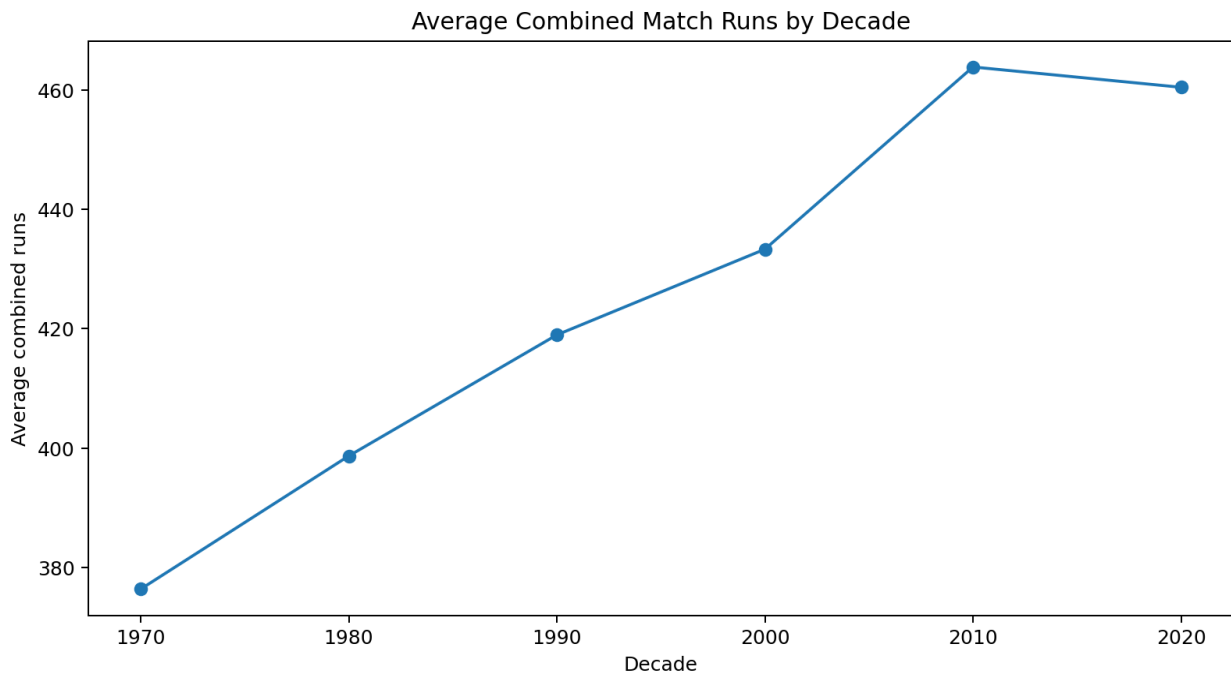


Figure 2. Average combined runs scored by both teams in a match by decade.

6. Toss and Match Outcome Analysis

Among 4,535 matches with a recorded toss winner, toss decision and match winner, the toss winner won the match in 50.34% of cases. This is close to 50%, suggesting that winning the toss alone is not a strong predictor of the match result in this dataset. The team batting first won 48.47% of the analyzed matches.

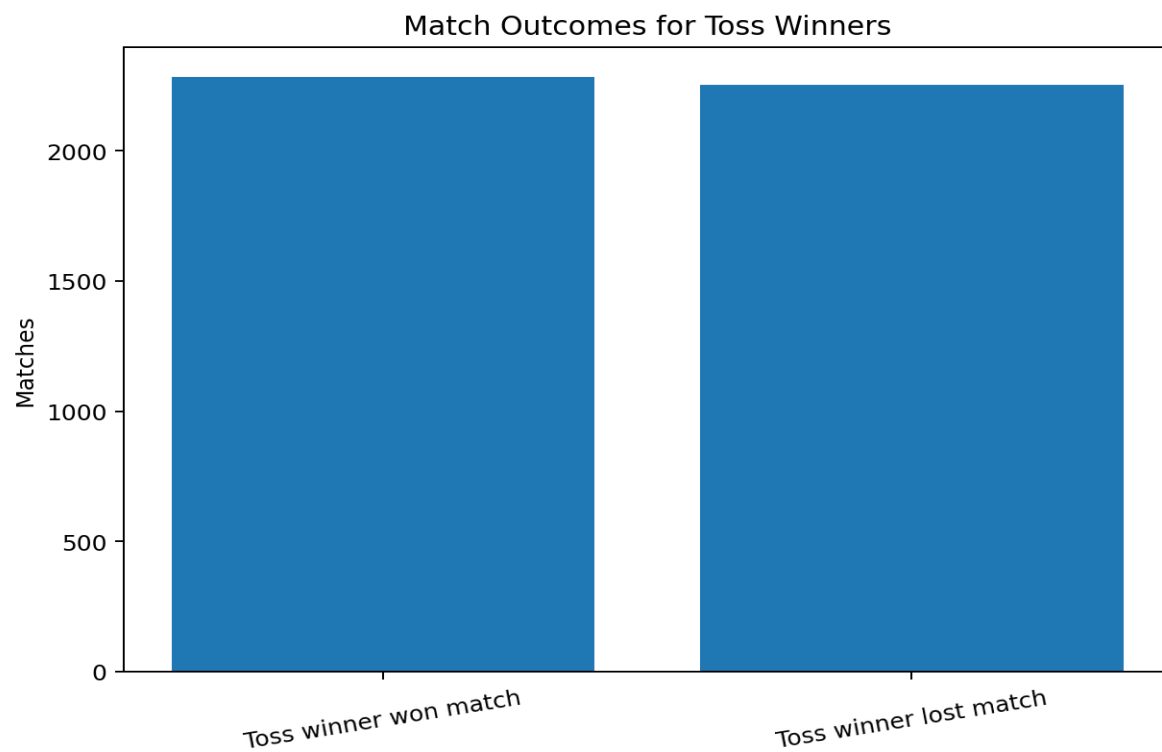


Figure 3. Match outcomes for teams that won the toss.

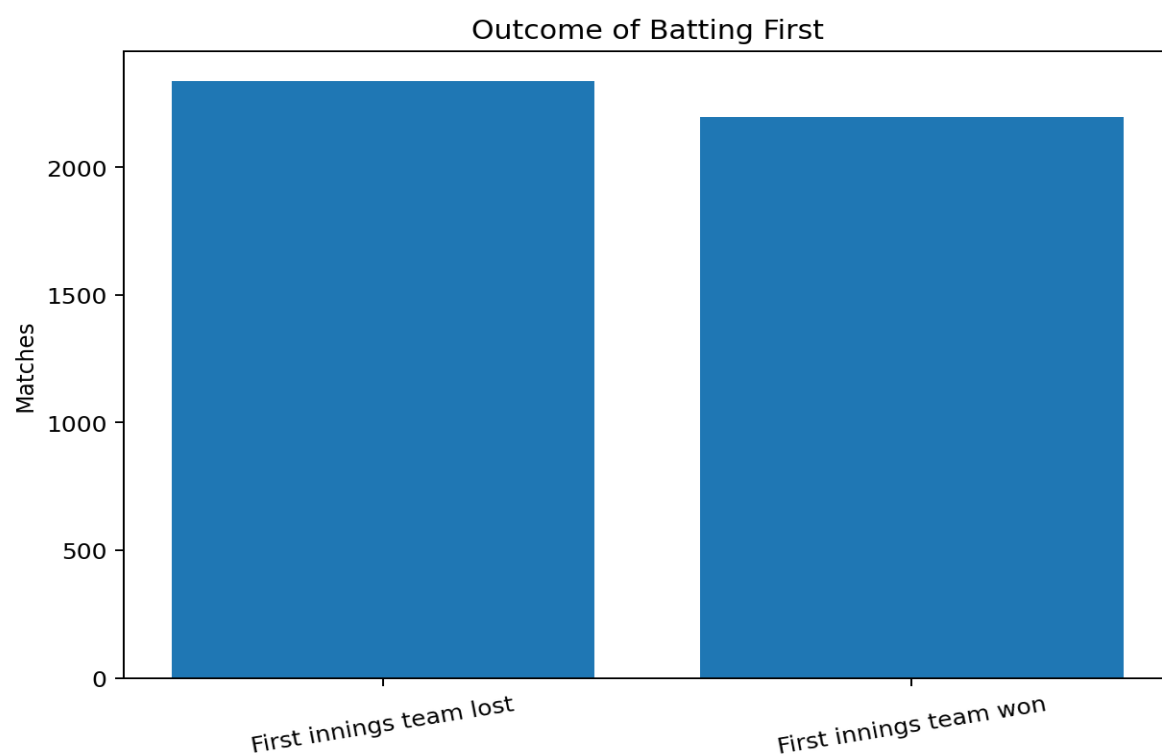


Figure 4. Outcomes for teams batting first.

7. Team Performance

Australia has the largest number of recorded wins in the dataset with 608 wins from 956 decisive matches, corresponding to a win rate of 63.60%. South Africa has a similarly high win rate among the major teams represented in the dataset. Win rate should be interpreted with caution because teams have played different numbers of matches and faced different opponents.

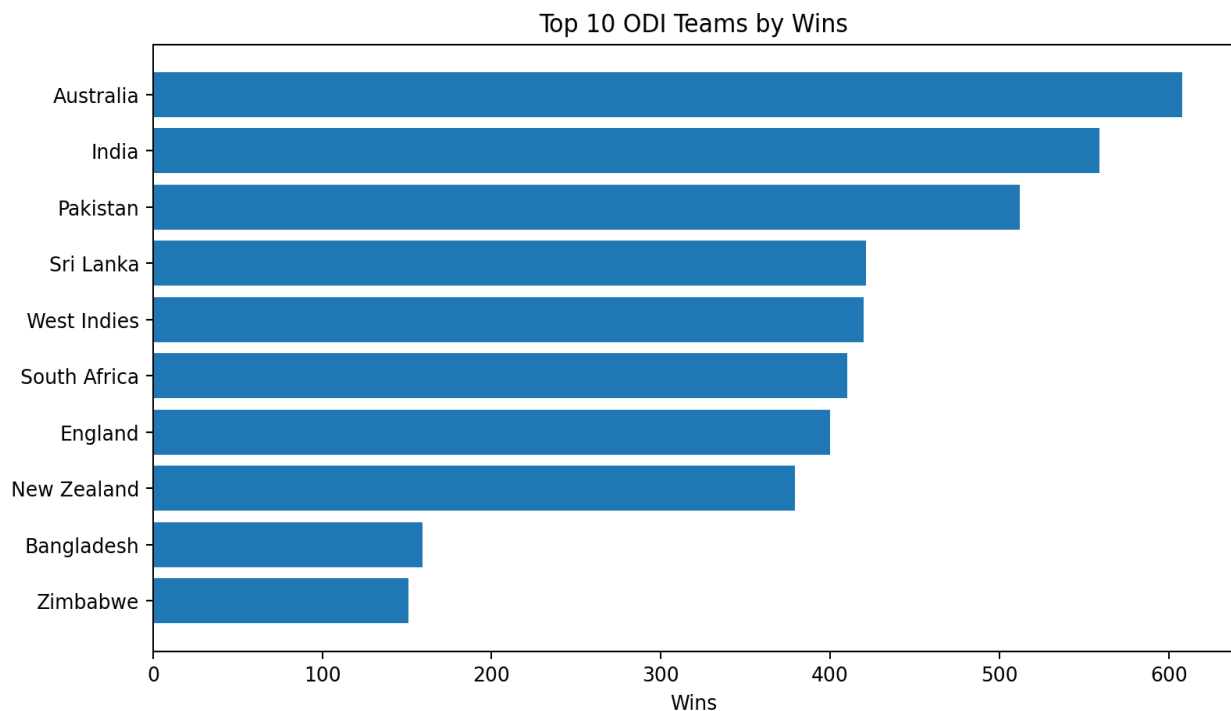


Figure 5. Top 10 teams by number of wins.

8. Individual Performance

The batting data identifies Sachin Tendulkar as the leading run scorer represented in the dataset with 18,426 runs. The bowling data identifies Muthiah Muralidaran as the leading wicket taker with 534 wickets.

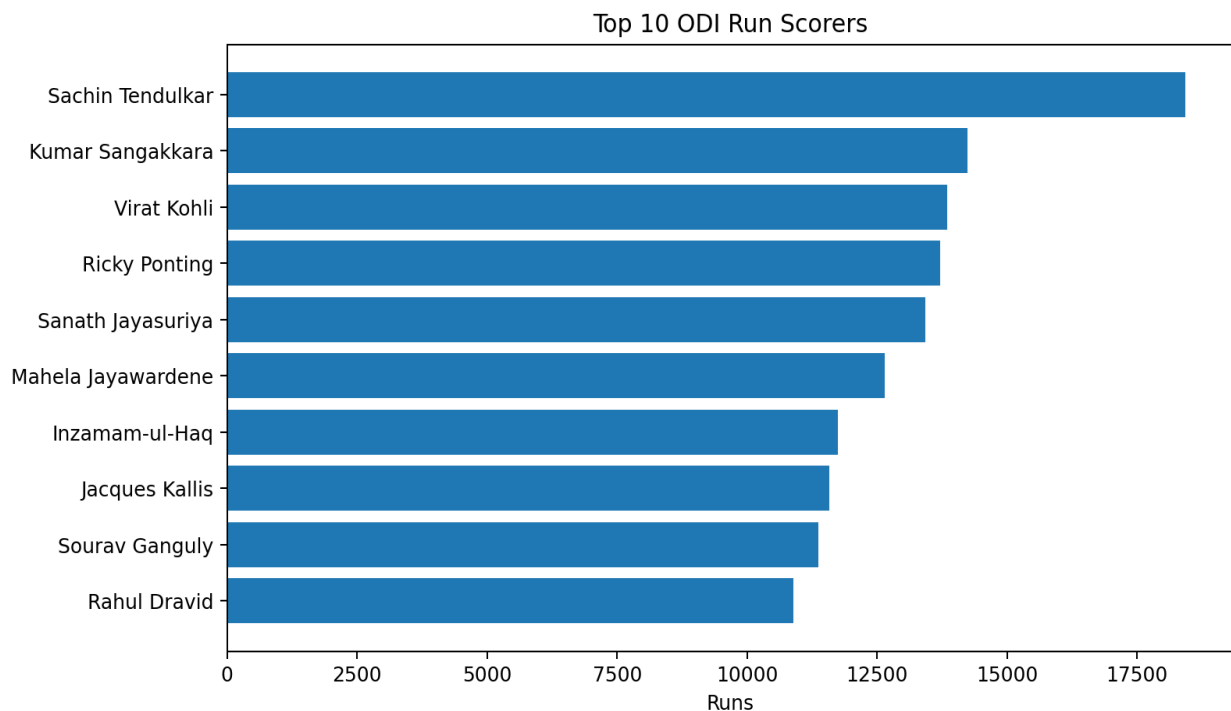


Figure 6. Top 10 run scorers represented in the batting data.

9. Key Findings

- ODI match coverage in the supplied dataset spans 1971-01-05 to 2024-03-18.
- The dataset contains 4,745 match records, with 4,535 having a recorded winner.
- Toss winners won 50.34% of analyzed matches, which is close to an even split.
- Teams batting first won 48.47% of analyzed matches.
- Australia has the highest number of wins (608).
- Sachin Tendulkar is the leading run scorer represented in the batting data (18,426 runs).
- Muthiah Muralidaran is the leading wicket taker represented in the bowling data (534 wickets).
- Average combined match scoring is higher in the 2000s and 2010s than in the earliest decades represented.

10. Conclusion

The analysis demonstrates how historical cricket data can be transformed into measurable insights using Python. ODI scoring has generally increased over time, while toss outcomes remain close to an even split with respect to match wins. Team and player aggregations reveal clear differences in long-term performance. The project also demonstrates the importance of data cleaning and of excluding incomplete records from calculations that depend on missing fields.

11. Limitations and Future Scope

The analysis is descriptive and does not establish causality. Toss advantage may depend on venue, weather, pitch conditions and team strength. Historical playing conditions and rules also changed over time. Future work could build predictive models for match outcomes, perform venue-specific analysis, compare eras, study player aging, or incorporate ball-by-ball data.

12. Tools and Technologies

Python, Pandas, NumPy and Matplotlib were used for data processing, aggregation and visualization.

13. Reference

Kaggle — ODI Cricket Matches Dataset (1971–2024), downloaded for the IBM SkillsBuild data analysis project.